CO5- AT3

K. JeeVan
192525138

Mock Interview

1) Suppose you are working on an Industrial monitoring system. How would you ~~see~~ use Bayesian Networks to predict machine failures.

Ans: A Bayesian Network represents relationship between machine conditions and failure. Variables like temperature, vibration and pressure are represented as Nodes. The model learns from previous data. New sensor data is used to calculate probability of failure. It helps in early fault detection and preventive maintainence.

2) Imagine you are asked in interview to design an Image denoising solution. How would markov Random Fields help model Neighbourhood pixel.

Ans: An markov Random field represents each image pixel as a node. Each pixel is connected to its neighbouring pixels. It uses neighbouring information to identify noisy pixels. The model estimates the most likely correct pixel values. This helps to remove noise while preserving important image details.

3) Suppose an interviewer asks how conditional independence simplifies probablistic interference. How would you justify it with graphical node.

Ans. Conditional independence means two variables become independent when another variable is known. For example, if disease causes fever and cough. If disease is known, the relation b/w fever and cough can be simplified. This reduces the number of probability calculations. It makes probablistic inference faster & easier.

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(4) In healthcare diagnosis problem, how would you Explain the use of comparision of directed and undirected graphical nodes for representing the symptoms relationship.

Ans, Directed models use arrows to represent the relationships b/w the variables. A Bayseian Network is best example. Undirected models use connections without arrows. A markov Random field is best example. Directed model shows directional relationship, while undirected model shows dependencies.